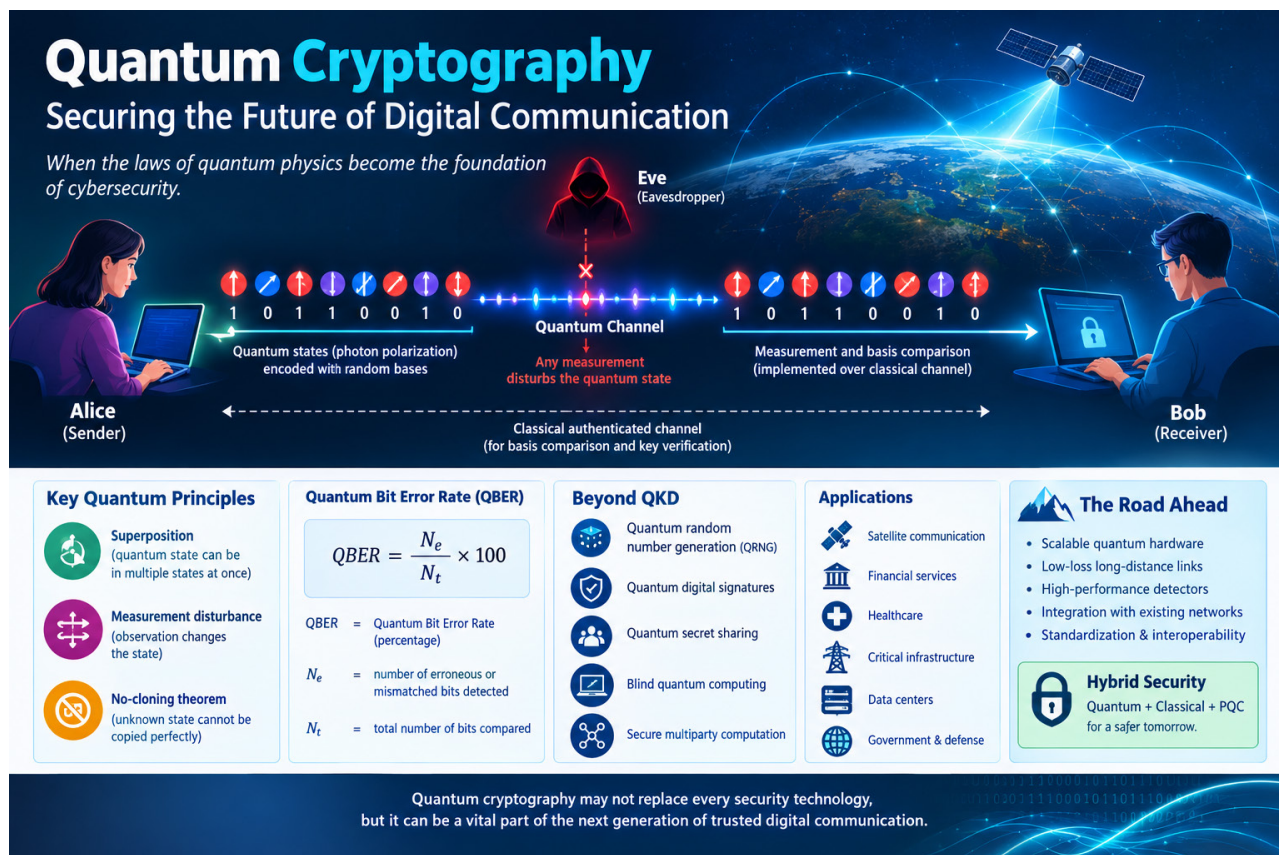


Quantum Cryptography: Ensuring Secure Digital Communication

The digital world is entering an era in which the security of information can no longer depend solely on traditional cryptographic systems. Quantum cryptography is poised to become a significant component of future digital communication.



From online banking and digital healthcare to satellite communication and critical infrastructure, enormous volumes of sensitive information are transmitted every second. Traditional cryptographic systems have protected this information for decades, but the emergence of quantum computing is challenging many of the assumptions on which these systems are built. Quantum cryptography offers a fundamentally different approach: instead of relying

only on mathematical problems that are difficult to solve, it uses the laws of quantum mechanics to detect eavesdropping and establish highly secure communication.

One of the most promising applications of quantum cryptography is Quantum Key Distribution (QKD). QKD allows two legitimate users to generate a shared secret cryptographic key while providing a mechanism to detect whether an attacker has attempted to intercept the communication.

From mathematical security to physical security

Most modern cryptographic systems use mathematical problems as the basis of security. For example, widely deployed public-key algorithms depend on the computational difficulty of problems such as integer factorization or discrete logarithms. A sufficiently powerful quantum computer, however, could use algorithms such as Shor's algorithm to solve some of these problems much more efficiently.

Quantum cryptography takes a different path. It exploits fundamental